# VideoFactory

## Complete Setup Guide for Mac and Windows

Your autonomous AI short-video factory: from topic to a fact-checked, narrated, captioned, quality-checked video, uploaded to YouTube through the official API, and controlled from your phone.

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# Contents


- 1. Before you start: what this guide gives you
- 2. What you need, and what it costs
- 3. Terminal and PowerShell basics (read this first)
- Part A: Mac setup, steps M1 to M12
- Part B: Windows setup, steps W1 to W12
- Part C: Google Cloud and YouTube permission (both computers)
- Part D: Phone setup (Samsung Galaxy S26 Ultra or any Android phone)
- Part E: Using the factory, from channel to YouTube
- Part F: Everyday commands
- Part G: Troubleshooting
- Part H: Security and backups
- Part I: What comes next
- Appendix: costs and limits, folder map, progress checklist

# 1. Before you start

This guide sets up **VideoFactory** from nothing. It is the clean, complete version of everything we built together, step by step, in the order that works. Follow it exactly and you will end with the same working factory you have now.

### What you will have at the end

- A web dashboard you open on your phone, protected by your own password.
- Channels with their own niche, audience, voice and visual style.
- AI topic ideas, scored and checked against past topics so nothing repeats.
- Research from real Wikipedia articles, with every fact checked against an exact quote from the source.
- A 45 to 55 second script, a scene plan, legally usable pictures with credits, an AI narrator, big captions, and a finished 1080×1920 MP4 video.
- An automatic quality check that fixes problems by itself, and asks you only when it must.
- Upload to YouTube through Google's official system, plus title, description and hashtags written for you.
- A factory that starts by itself, stays awake, and is reachable from your phone anywhere through a private, encrypted link.

### Which part should you follow?

| Your situation | Follow |
|---|---|
| Setting up on a Mac | Part A, then Part C, Part D, Part E |
| Setting up on a Windows PC | Part B, then Part C, Part D, Part E |
| Your Mac is already working (built during our chat) | Nothing to reinstall. Keep this as your reference. Parts E to H are useful every day. |

**HONEST NOTE:** The Mac steps are exactly what we ran and tested together on your 2019 Intel MacBook Pro with macOS Tahoe. The Windows steps are careful equivalents, and the app code includes the changes Windows needs, but it has not yet been run on a real Windows PC. If any Windows step shows an error, copy the exact message and send it to your AI co-founder; it will be fixed step by step like everything else.

### How each step is written

Every step tells you the **device**, the **cost**, what you are doing and why, the exact thing to **copy**, the **expected result**, and what to do **if it fails**. Do one step at a time. Never move on while a step shows an error.

# 2. What you need, and what it costs

## Devices

| Device | Used for |
|---|---|
| Mac or Windows computer | Runs the factory: AI requests, FFmpeg video building, the database and the dashboard. Must stay switched on, awake and plugged in while making videos. |
| Android phone (S26 Ultra) | Your control centre: dashboard, watching videos, starting jobs, approving, uploading. |

## Minimum computer

- Mac: macOS 12 or newer, Intel or Apple chip, 8 GB memory (16 GB is better), 20 GB free storage.
- Windows: Windows 10 or 11 (64-bit), 8 GB memory, 20 GB free storage.
- A home internet connection.

## Accounts (all free)

| Account | Why | Cost |
|---|---|---|
| Google account that owns a YouTube channel | AI (Gemini), Google Cloud, YouTube | Free |
| Pixabay account (optional) | Extra free stock photos | Free |
| Tailscale (sign in with Google) | Permanent private link to your factory from anywhere | Free |

> **COST:** Everything in this guide is free. Never add a credit card or billing in Google Cloud or AI Studio: without billing, Google cannot charge you. The free services have daily limits; when a limit is reached the factory simply waits and tries again later. Full details are in the Appendix.

# 3. Terminal and PowerShell basics

Some steps need you to type instructions to the computer. You will only ever **paste lines from this guide**. Nothing here can harm your computer.

### On a Mac: Terminal

1. Press **Command + Space**, type **Terminal**, press **Return**.
2. A window opens with a line ending in **%**. That means it is ready.
3. Copy a command from this guide, click in Terminal, press **Command + V** to paste, then **Return** to run it.
4. Wait until the **%** line comes back before pasting the next command.
5. To stop something that keeps running, press **Control + C**.

### On Windows: PowerShell

1. Click **Start**, type **PowerShell**, click **Windows PowerShell** (or **Terminal**).
2. A window opens with a line ending in **>**. That means it is ready.
3. Copy a command, click in the window, press **Ctrl + V** (or right-click) to paste, then **Enter**.
4. Wait until the **>** line comes back before the next command.
5. To stop something that keeps running, press **Ctrl + C**.

> **PASSWORDS ARE INVISIBLE:** When a command asks for a password or a key, nothing appears while you type or paste: no dots, no stars. This is a security feature. Type or paste carefully, then press Return or Enter.
>
> **SECRETS:** Never paste your keys, passwords or the YouTube client file into a chat, email or website. Nobody helping you ever needs to see them.

# Part A: Mac setup

Steps M1 to M12. Use Terminal for the commands and Chrome for the websites.

## M1. Check what your Mac can handle

**DEVICE:** Mac **COST:** Free **TIME:** 2 minutes

**Why:** the computer's chip and memory decide which tools can run for free.

1. Click the **Apple logo** (top left) and choose **About This Mac**. Note **Chip/Processor**, **Memory** and **macOS**.
2. Apple logo, then **System Settings**, **General**, **Storage**. Note the free space.

**EXPECTED RESULT:** You meet the minimum in section 2. An Intel Mac works fine: the factory uses Google's free AI instead of running AI on the Mac.

## M2. Install Python 3.13 (official version)

**DEVICE:** Mac **COST:** Free (open source) **TIME:** 5 minutes

**What:** Python is the language the factory is written in. We install the official version from python.org, not Homebrew's, because Homebrew is reducing its support for Intel Macs.

**1. Download the installer:**

```
cd ~/Downloads && curl -L -o python-3.13.16-macos11.pkg
https://www.python.org/ftp/python/3.13.16/python-3.13.16-macos11.pkg
```

**2. Install it** (asks for your Mac password; typing is invisible):

```
sudo installer -pkg python-3.13.16-macos11.pkg -target /
```

**3. Allow secure downloads:**

```
sh "/Applications/Python 3.13/Install Certificates.command"
```

**4. Check:**

```
ls /Library/Frameworks/Python.framework/Versions/ &&
/Library/Frameworks/Python.framework/Versions/3.13/bin/python3.13 --version
```

**EXPECTED RESULT:** 3.13 Current, then Python 3.13.16 (a newer 3.13.x number is also fine; if python.org shows a newer 3.13 version, use its link instead).

**IF IT FAILS:**

- “Sorry, try again”: the password was mistyped. Run command 2 again.
- “No such file or directory” in step 4: the installer did not finish. Repeat steps 1 and 2 and send the exact message if it happens again.

## M3. Unpack the VideoFactory package

**DEVICE:** Mac **COST:** Free **TIME:** 2 minutes

**What:** the package **VideoFactory-complete.zip** contains all the app code. It contains no keys, passwords or videos. Download it into your **Downloads** folder first.

```
cd ~ && unzip -o ~/Downloads/VideoFactory-complete.zip -d ~ && chmod +x
~/VideoFactory/factory.sh ~/VideoFactory/start.sh && ls ~/VideoFactory
```

**EXPECTED RESULT:** A list including app, bin, factory.sh, requirements.txt, start.sh.

**ALREADY SET UP?:** If your Mac already runs VideoFactory, unpacking again only replaces app code. Your database, videos, keys, password and YouTube permission (in data, .env and secrets) are never touched. Afterwards run: bash ~/VideoFactory/factory.sh restart

## M4. Install FFmpeg (the video builder)

**DEVICE:** Mac **COST:** Free (open source) **TIME:** 3 minutes

**What:** FFmpeg turns pictures, voice and captions into the final video. FFprobe inspects videos for the quality check. We use ready-made Intel-compatible builds that can draw captions.

```
cd ~/VideoFactory/bin && curl -L -o ffmpeg.zip
https://evermeet.cx/ffmpeg/getrelease/ffmpeg/zip && curl -L -o ffprobe.zip
https://evermeet.cx/ffmpeg/getrelease/ffprobe/zip

unzip -o ffmpeg.zip && unzip -o ffprobe.zip && rm ffmpeg.zip ffprobe.zip && ./ffmpeg -
hide_banner -filters | grep -E "subtitles|drawtext"
```

**EXPECTED RESULT:** Lines mentioning drawtext and subtitles. These are what put captions on your videos.

**IF IT FAILS:**

- “End-of-central-directory signature not found”: the download site sent the wrong file. Ask your AI co-founder for a backup download source.
- A popup says ffmpeg cannot be opened: click Done (not Move to Trash) and send a screenshot.

## M5. Create the factory's private toolbox

**DEVICE:** Mac **COST:** Free **TIME:** 3 minutes

**What:** a “virtual environment” is a private toolbox just for this project, so its add-ons never mix with the rest of your Mac.

```
cd ~/VideoFactory && /Library/Frameworks/Python.framework/Versions/3.13/bin/python3.13 -m
venv venv && venv/bin/python -m pip install --upgrade pip && venv/bin/python -m pip
install -r requirements.txt
```

**EXPECTED RESULT:** It ends with “Successfully installed …” listing flask, waitress and tzdata.

## M6. Free AI key (Google Gemini)

**DEVICE:** Mac (Chrome + Terminal) **COST:** Free tier, no card **TIME:** 5 minutes

1. In Chrome go to **https://aistudio.google.com/apikey** and sign in with your Google account.
2. Click **Create API key** (choose or create a project if asked). Copy the key that starts with **AIza**.
3. **Do not** add billing anywhere.

Save it safely (paste when asked; it stays invisible):

```
cd ~/VideoFactory && venv/bin/python -m app.set_key GEMINI_API_KEY
```

**EXPECTED RESULT:** GEMINI_API_KEY SAVED SAFELY (first 4 characters: AIza...)

> **PRIVACY:** On Google's free tier, what you send may be used to improve Google's models and may be read by reviewers. That is fine for public history facts. Never put private information into the factory.

## M7. Choose your dashboard password

**DEVICE:** Mac **COST:** Free **TIME:** 1 minute

At least 10 characters, new (not your Mac or Google password). Typing is invisible.

```
cd ~/VideoFactory && venv/bin/python -m app.set_password
```

**EXPECTED RESULT:** PASSWORD SAVED.

## M8. First start and system check

**DEVICE:** Mac **COST:** Free **TIME:** 5 minutes

```
cd ~/VideoFactory && venv/bin/waitress-serve --listen=127.0.0.1:8000 app.main:app
```

1. Leave Terminal running. In Chrome open **http://127.0.0.1:8000** and log in.
2. Tap **Open system check**, then **Test AI connection**. Wait up to 20 seconds.
3. Check the cards. YouTube will say “Not set up” until Part C; that is expected.
4. Back in Terminal, press **Control + C** to stop this test run.

**EXPECTED RESULT:** Green dots for Python, FFmpeg, FFprobe, Caption drawing, Database, AI brain and disk space.

## M9. Start automatically and stay awake

**DEVICE:** Mac **COST:** Free (built into macOS) **TIME:** 2 minutes

**What:** macOS will run the factory in the background, start it every time you log in, restart it if it crashes, and keep the Mac awake while it is plugged in.

```
bash ~/VideoFactory/factory.sh install
```

**EXPECTED RESULT:** INSTALLED AND RUNNING, followed by a phone link.

If macOS asks whether **Python** may accept incoming network connections, click **Allow**. You can now quit Terminal; the factory keeps running.

> **KEEP THE LID OPEN:** A MacBook always sleeps when its lid is closed. Keep it open and plugged in. You can turn the brightness all the way down.

## M10. Permanent private link (Tailscale)

**DEVICE:** Mac **COST:** Free (Personal plan) **TIME:** 10 minutes

**What:** Tailscale gives your Mac a private address starting with **100.** that never changes and works from anywhere, only for your own devices. Do not also install the App Store version.

```
cd ~/Downloads && curl -L -o Tailscale-latest-macos.pkg
https://pkgs.tailscale.com/stable/Tailscale-latest-macos.pkg && sudo installer -pkg
Tailscale-latest-macos.pkg -target / && rm Tailscale-latest-macos.pkg && open -a
Tailscale
```

1. When macOS asks, allow the Tailscale **network extension** (System Settings) and **VPN configuration**.
2. Log in with your **Google account** in the browser page that opens. Use the same account on your phone later.

```
bash ~/VideoFactory/factory.sh status
```

**EXPECTED RESULT:** STATUS: RUNNING and PHONE LINK (works anywhere, never changes): http://100.x.y.z:8000. Write that link down.

## M11. Extra free pictures (Pixabay, optional)

**DEVICE:** Mac (Chrome + Terminal) **COST:** Free **TIME:** 5 minutes

Without this key the factory uses Wikimedia Commons only. Pexels was the first choice, but it paused new keys in October 2026.

1. In Chrome open **https://pixabay.com/api/docs/**, create a free account and log in.
2. On the same page, under **Parameters**, copy the green value next to **key (required)**.

```
cd ~/VideoFactory && venv/bin/python -m app.set_key PIXABAY_API_KEY && bash factory.sh
restart
```

**EXPECTED RESULT:** PIXABAY_API_KEY SAVED SAFELY, then STATUS: RUNNING.

## M12. YouTube permission

**DEVICE:** Mac **COST:** Free **TIME:** 15 minutes

First do **Part C** (Google Cloud, in Chrome). Then store the downloaded client file. This takes the newest **client_secret** file from Downloads and checks it:

```
mkdir -p ~/VideoFactory/secrets && chmod 700 ~/VideoFactory/secrets && mv "$(ls -t
~/Downloads/client_secret_*.json | head -n 1)" ~/VideoFactory/secrets/youtube_client.json
&& chmod 600 ~/VideoFactory/secrets/youtube_client.json && ~/VideoFactory/venv/bin/python
-c "import
json;d=json.load(open('$HOME/VideoFactory/secrets/youtube_client.json'));print('YOUTUBE
CLIENT SAVED. Type:', list(d)[0])"
```

**EXPECTED RESULT:** YOUTUBE CLIENT SAVED. Type: installed

Then connect your channel: in Chrome **on the Mac**, open **http://127.0.0.1:8000/youtube**, log in, tap **Connect YouTube**, and follow C6 in Part C.

**CAREFUL:** Delete any old client_secret files from Downloads only by hand in Finder, and only after you are sure they are not used by another project.

# Part B: Windows setup

Steps W1 to W12. Use **PowerShell** for the commands (see section 3) and Chrome or Edge for the websites. Paths like **$HOME** mean your own user folder, for example C:\Users\YourName.

## W1. Check what your PC can handle

**DEVICE:** Windows **COST:** Free **TIME:** 2 minutes

1. Open **Settings**, **System**, **About**. Note **Processor**, **Installed RAM** and **System type** (must be 64-bit).
2. **Settings**, **System**, **Storage**: note the free space on drive C.

**EXPECTED RESULT:** You meet the minimum in section 2.

## W2. Install Python 3.13

**DEVICE:** Windows **COST:** Free (open source) **TIME:** 5 minutes

```
winget install --id Python.Python.3.13 -e --accept-package-agreements --accept-source-agreements
```

**Close PowerShell and open a new one** (so it notices Python), then check:

```
py -3.13 --version
```

**EXPECTED RESULT:** Python 3.13.x

**IF IT FAILS:**

- “winget is not recognized”: download the installer instead. Go to https://www.python.org/downloads/windows/, choose the latest Python 3.13 “Windows installer (64-bit)”, run it, tick **Add python.exe to PATH**, then click **Install Now**.
- “py is not recognized” after installing: close every PowerShell window, open a new one, and try again.

## W3. Unpack the VideoFactory package

**DEVICE:** Windows **COST:** Free **TIME:** 2 minutes

Download **VideoFactory-complete.zip** into your **Downloads** folder, then:

```
Expand-Archive -Path "$HOME\Downloads\VideoFactory-complete.zip" -DestinationPath "$HOME" -Force; Get-ChildItem "$HOME\VideoFactory"
```

**EXPECTED RESULT:** A list including app, bin, requirements.txt and start-windows.bat.

## W4. Install FFmpeg (the video builder)

**DEVICE:** Windows **COST:** Free (open source) **TIME:** 5 minutes

We use the widely used Windows “essentials” build, which can draw captions.

```
$ProgressPreference = 'SilentlyContinue'
Invoke-WebRequest -Uri "https://www.gyan.dev/ffmpeg/builds/ffmpeg-release-essentials.zip"
-OutFile "$env:TEMP\ffmpeg.zip"
Expand-Archive -Path "$env:TEMP\ffmpeg.zip" -DestinationPath "$env:TEMP\ffmpeg" -Force
Get-ChildItem "$env:TEMP\ffmpeg" -Recurse -Include ffmpeg.exe,ffprobe.exe | Copy-Item -
Destination "$HOME\VideoFactory\bin" -Force
& "$HOME\VideoFactory\bin\ffmpeg.exe" -hide_banner -filters | Select-String "subtitles|
drawtext"
```

**EXPECTED RESULT:** Lines mentioning drawtext and subtitles.

**IF IT FAILS:**

- No lines appear: that build cannot draw captions. Replace the download address in the second line with https://github.com/BtbN/FFmpeg-Builds/releases/download/latest/ffmpeg-master-latest-win64-gpl.zip and run all five lines again.

## W5. Create the factory's private toolbox

**DEVICE:** Windows **COST:** Free **TIME:** 3 minutes

```
cd "$HOME\VideoFactory"
py -3.13 -m venv venv
.\venv\Scripts\python.exe -m pip install --upgrade pip
.\venv\Scripts\python.exe -m pip install -r requirements.txt
```

**EXPECTED RESULT:** “Successfully installed …” listing flask, waitress and tzdata.

## W6. Free AI key (Google Gemini)

**DEVICE:** Windows (browser + PowerShell) **COST:** Free tier, no card **TIME:** 5 minutes

Create the key exactly as in **M6** (https://aistudio.google.com/apikey, no billing), then save it. Paste with right-click or Ctrl + V; it stays invisible.

```
cd "$HOME\VideoFactory"; .\venv\Scripts\python.exe -m app.set_key GEMINI_API_KEY
```

**EXPECTED RESULT:** GEMINI_API_KEY SAVED SAFELY (first 4 characters: AIza...)

## W7. Choose your dashboard password

**DEVICE:** Windows **COST:** Free **TIME:** 1 minute

```
cd "$HOME\VideoFactory"; .\venv\Scripts\python.exe -m app.set_password
```

**EXPECTED RESULT:** PASSWORD SAVED.

## W8. First start and system check

**DEVICE:** Windows **COST:** Free **TIME:** 5 minutes

```
cd "$HOME\VideoFactory"; .\venv\Scripts\waitress-serve.exe --listen=127.0.0.1:8000
app.main:app
```

Open **http://127.0.0.1:8000** in your browser, log in, open the system check and tap **Test AI connection** (same as M8). Then press **Ctrl + C** in PowerShell.

**TIME ZONE:** On Windows the channel form cannot read your time zone automatically. When you create a channel, type it yourself in the standard form Region/City, for example Europe/London or America/New_York.

## W9. Start automatically and stay awake

**DEVICE:** Windows **COST:** Free (built into Windows) **TIME:** 5 minutes

**1. Create the automatic start** (runs at every login, in a minimized window):

```
$bat = "$HOME\VideoFactory\start-windows.bat"
$action = New-ScheduledTaskAction -Execute "cmd.exe" -Argument "/c start `"VideoFactory`"
/min `"$bat`""
$trigger = New-ScheduledTaskTrigger -AtLogOn -User $env:USERNAME
$settings = New-ScheduledTaskSettingsSet -AllowStartIfOnBatteries -
DontStopIfGoingOnBatteries -ExecutionTimeLimit ([TimeSpan]::Zero)
Register-ScheduledTask -TaskName "VideoFactory" -Action $action -Trigger $trigger -
Settings $settings -Force
Start-ScheduledTask -TaskName "VideoFactory"
```

If Windows Security asks whether Python may communicate on networks, tick **Private networks** and click **Allow**.

**2. Never sleep while plugged in:**

```
powercfg /change standby-timeout-ac 0
```

**3. Optional, laptops: keep working with the lid closed while plugged in:**

```
powercfg /setacvalueindex SCHEME_CURRENT SUB_BUTTONS LIDACTION 0; powercfg /setactive
SCHEME_CURRENT
```

**4. Check it is running:**

```
(Invoke-WebRequest http://127.0.0.1:8000/login -UseBasicParsing).StatusCode
```

**EXPECTED RESULT:** 200. A minimized “VideoFactory” window appears in the taskbar. Do not close it; closing it stops the factory until the next login.

**IF IT FAILS:**

- “Access is denied” when registering the task: right-click PowerShell, choose **Run as administrator**, and run part 1 again.
- Your phone cannot reach the PC on home Wi-Fi: open Settings, Network & internet, Wi-Fi, your network, and set it to **Private network**.

## W10. Permanent private link (Tailscale)

**DEVICE:** Windows **COST:** Free (Personal plan) **TIME:** 10 minutes

```
winget install --id Tailscale.Tailscale -e --accept-package-agreements --accept-source-
agreements
```

Open **Tailscale** from the Start menu (or the tray icon near the clock), click **Log in**, and sign in with your **Google account**. Then:

```
tailscale ip -4
```

**EXPECTED RESULT:** An address starting with 100. Your permanent phone link is http://100.x.y.z:8000

**IF IT FAILS:**

- “tailscale is not recognized”: close PowerShell, open a new one and try again, or read the 100. address from the Tailscale tray icon.

## W11. Extra free pictures (Pixabay, optional)

**DEVICE:** Windows (browser + PowerShell) **COST:** Free **TIME:** 5 minutes

Get the key as in **M11**, then save it and restart:

```
cd "$HOME\VideoFactory"; .\venv\Scripts\python.exe -m app.set_key PIXABAY_API_KEY
taskkill /IM waitress-serve.exe /F; Start-ScheduledTask -TaskName "VideoFactory"
```

**EXPECTED RESULT:** PIXABAY_API_KEY SAVED SAFELY.

## W12. YouTube permission

**DEVICE:** Windows **COST:** Free **TIME:** 15 minutes

First do **Part C**. Then store the newest downloaded client file and check it:

```
New-Item -ItemType Directory -Force "$HOME\VideoFactory\secrets" | Out-Null
$f = Get-ChildItem "$HOME\Downloads\client_secret_*.json" | Sort-Object LastWriteTime -
Descending | Select-Object -First 1
Move-Item $f.FullName "$HOME\VideoFactory\secrets\youtube_client.json" -Force
cd "$HOME\VideoFactory"; .\venv\Scripts\python.exe -c "import
json;d=json.load(open('secrets/youtube_client.json'));print('YOUTUBE CLIENT SAVED.
Type:', list(d)[0])"
```

**EXPECTED RESULT:** YOUTUBE CLIENT SAVED. Type: installed

Then, on the PC, open **http://127.0.0.1:8000/youtube**, log in, tap **Connect YouTube** and follow C6.

# Part C: Google Cloud and YouTube permission

**DEVICE:** Mac or Windows (browser) **COST:** Free: never add billing **TIME:** 20 minutes

**What:** you register your factory with Google as an app allowed to ask for YouTube access. You then approve it once for your own channel. Your Google password is never given to the factory.

> **BEFORE YOU START:** You need a YouTube channel on the Google account you use. In the YouTube app: profile picture, then Create a channel.
>
> **TWO FACTS:** 1) The YouTube API is free with a daily allowance, enough for several uploads a day. 2) YouTube keeps videos uploaded by new, unreviewed apps PRIVATE until the app passes a free YouTube review (an “audit”). Test uploads work; they are just private, and you can make them public yourself in YouTube Studio.

## C1. Create a project

1. Open **https://console.cloud.google.com/** and sign in with the Google account that owns your channel.
2. Click the **project selector** at the top left, then **New project**. Name: **VideoFactory**. Click **Create**.
3. Make sure **VideoFactory** is the selected project at the top.

## C2. Switch on the YouTube APIs

1. Open **https://console.cloud.google.com/apis/library**
2. Search **YouTube Data API v3**, open it, click **Enable**.
3. Search **YouTube Analytics API**, open it, click **Enable** (used later for view counts).

## C3. Set up Google Auth (consent screen)

1. Open **https://console.cloud.google.com/auth/overview** and click **Get started**.
2. **App Information:** App name **VideoFactory**. For User support email, **click** your email in the list (do not type it). Next.
3. **Audience: External**. Next.
4. **Contact information:** type your email and press **Return/Enter** so it becomes a small label. Next.
5. Tick **I agree**, then **Continue** and **Create**.

**EXPECTED RESULT:** The page shows “OAuth Overview”. That means branding is saved.

> **DO NOT:** Do not upload a logo and do not fill in website, privacy policy or authorized domain fields. A logo forces Google's verification review.

## C4. Add yourself as a test user

1. In the left menu click **Audience**.
2. Under **Test users**, click **+ Add users**, type the Gmail address that owns your channel, click **Save**.

**EXPECTED RESULT:** Your email is listed under Test users.

**WHY TESTING MODE:** Publishing the app to “In production” needs more branding details (a website and privacy policy), so the Publish app button stays grey. Testing mode works now. The only difference: Google ends the YouTube connection every 7 days, so you reconnect once a week (two clicks). The dashboard warns you a day before.

## C5. Create the factory's ID card and download it

1. In the left menu click **Clients**, then **+ Create client** (or **Create OAuth client** on the Overview page).
2. Application type: **Desktop app**. Name: **VideoFactory Mac** (or PC). Click **Create**.
3. Click **Download JSON**. A file named client_secret_….json appears in Downloads.
4. Go back to **M12** (Mac) or **W12** (Windows) to store it.

**SECRET:** This file is the key to your factory's YouTube access. Do not share it or send it to anyone.

## C6. Connect your channel (on the computer)

1. On the computer itself, open **http://127.0.0.1:8000/youtube** (it must be 127.0.0.1, not the 100. link) and log in.
2. Tap **Connect YouTube** and choose the Gmail account you added as a test user.
3. Google shows **“Google hasn't verified this app”**. This is expected for your own app. Click **Continue** (or **Advanced**, then **Go to VideoFactory**).
4. **Tick every permission box** (or Select all), then **Continue**. If asked, choose the channel to upload to.

**EXPECTED RESULT:** Back on VideoFactory: “YouTube connected: (your channel)” and a CONNECTED badge.

**IF IT FAILS:**

- “Access blocked” or “has not completed the Google verification process”: the account is not a test user. Do C4, then connect again.
- “This Google account has no YouTube channel yet”: create a channel in the YouTube app, then connect again.
- “redirect_uri_mismatch”: the client is not a Desktop app. Redo C5 choosing Desktop app.

# Part D: Phone setup

**DEVICE:** Android phone (S26 Ultra) **COST:** Free **TIME:** 10 minutes

## D1. Tailscale on the phone

1. Open the **Play Store**, search **Tailscale**, tap **Install**, then **Open**.
2. Tap **Get started** and sign in with the **same Google account** as on the computer.
3. When Android asks to set up a VPN connection, tap **OK**.

**EXPECTED RESULT:** The app says Connected and lists your computer.

> **ONE VPN AT A TIME:** Android allows only one VPN app at a time. If you use another VPN, turn it off while using VideoFactory. Tailscale only carries traffic to your own devices; your normal internet is unaffected.

## D2. Open your dashboard

1. In Chrome, open your permanent link **http://100.x.y.z:8000** (from M10 or W10).
2. Log in. Let Samsung Pass or Google Password Manager save the password if offered.
3. Tap ⋮, then **Add to Home screen**, then **Add**. VideoFactory now opens like an app.
4. Test it away from home: turn Wi-Fi off and open the link over mobile data.

**EXPECTED RESULT:** Your dashboard opens over mobile data.

# Part E: Using the factory

## E1. Create a channel

1. On the dashboard tap **+ Create channel**.
2. Fill in **Channel name** and **Niche** (be specific, without quotation marks), for example: Little-known true stories from ancient civilisations.
3. Optional: audience, language, **narrator voice**, visual style, content rules (for example: No gore. Say clearly when something is a legend.), videos per day, time zone.
4. Tap **Save channel**. Channels start PAUSED until full autopilot is built.

## E2. Find and choose a topic

1. On the channel card tap **Topics**, then **Find new topic ideas**. Wait 20 to 60 seconds.
2. About 8 ideas appear, each scored out of 100. **BEST PICK** marks the top eligible idea. Tap **Score details** to see why.
3. Tap **Use this topic** on the one you like, or **Reject** to rule an idea out for good.

## E3. Make the video

1. On the **SELECTED** topic tap **Start video**.
2. The job page refreshes by itself and shows the production line: Research, Fact-check, Script, Scene plan, Visuals, Voice, Captions, Video assembly, Quality check.
3. It takes about 5 to 15 minutes. You can leave and come back; the work continues on the computer.
4. When it says **Ready**, watch the video, read the quality check, and copy the publishing text if you like. **Download video** saves the MP4 to your phone.

| What you see | What it means | What to do |
|---|---|---|
| RETRYING | A free limit or a busy service. The factory waits 1, 5, then 15 minutes and tries again. | Nothing. It continues by itself. |
| Working on: Script (after Quality check) | The quality check found a problem and is fixing it automatically (at most 2 rounds). | Nothing. This is self-correction, not an error. |
| NEEDS ATTENTION | Something retrying will not fix. | Read What happened, Why and Steps on the job page. Use Retry, Approve anyway (quality check only) or Cancel. |
| Ready | Passed quality control and saved on the computer. | Watch, then publish. |

## E4. Upload to YouTube

1. Open a **Ready** video. In **Publish**, tap **Upload to YouTube (private)**.
2. Wait 1 to 3 minutes. The banner shows **On YouTube (PRIVATE)** with a link.
3. To make it public later, open **YouTube Studio**, **Content**, the video, **Visibility**, **Public**.

**NEVER TWICE:** The factory records every upload and refuses to upload the same video twice.

## E5. Weekly YouTube reconnect (Testing mode)

When the dashboard shows **RECONNECT SOON** or **RECONNECT NEEDED**: on the computer, open **http://127.0.0.1:8000/youtube**, tap **Reconnect YouTube**, approve. Any upload that was waiting can then be retried from its job page.

# Part F: Everyday commands

## Mac (Terminal)

| What you want | Command |
|---|---|
| Is it running? Phone links | bash ~/VideoFactory/factory.sh status |
| Restart (after an update or new key) | bash ~/VideoFactory/factory.sh restart |
| Stop | bash ~/VideoFactory/factory.sh stop |
| Start again | bash ~/VideoFactory/factory.sh start |
| Recent activity and errors | bash ~/VideoFactory/factory.sh logs |
| Remove the automatic start | bash ~/VideoFactory/factory.sh uninstall |
| Install an update zip | cd ~/VideoFactory && unzip -o ~/Downloads/UPDATE.zip -d ~/VideoFactory && bash factory.sh restart |

## Windows (PowerShell)

| What you want | Command |
|---|---|
| Is it running? (200 = yes) | (Invoke-WebRequest http://127.0.0.1:8000/login -UseBasicParsing).StatusCode |
| Stop | taskkill /IM waitress-serve.exe /F |
| Start again | Start-ScheduledTask -TaskName "VideoFactory" |
| Recent activity and errors | Get-Content "$HOME\VideoFactory\data\logs\app.log" -Tail 40 |
| Permanent phone link | tailscale ip -4  (then http://ADDRESS:8000) |
| Home Wi-Fi address | ipconfig  (look for IPv4 Address) |
| Remove the automatic start | Unregister-ScheduledTask -TaskName "VideoFactory" -Confirm:$false |

### Where things are

| Item | Mac | Windows |
|---|---|---|
| Dashboard on the computer | http://127.0.0.1:8000 | http://127.0.0.1:8000 |
| Dashboard on the phone | http://100.x.y.z:8000 (Tailscale) | http://100.x.y.z:8000 (Tailscale) |
| Finished videos | ~/VideoFactory/data/jobs/NUMBER/video/final.mp4 | %USERPROFILE%\VideoFactory\data\jobs\NUMBER\video\final.mp4 |

# Part G: Troubleshooting

These are the real problems we met while building, and their fixes. For anything else: copy the exact error text (or take a screenshot) and send it to your AI co-founder. Never guess.

| Problem | Why | Fix |
|---|---|---|
| Phone says “This site can't be reached” | The factory is not running, the computer is asleep, or you used an old home-Wi-Fi address | Use the 100. link. On the Mac run factory.sh status. Keep the computer awake and plugged in. |
| Home address changed (for example .1.11 became .1.6) | Routers hand out new addresses | Always use the permanent Tailscale 100. link. |
| Terminal shows heredoc> lines | A long pasted block was cut off | Press Control + C. Use downloaded files instead of long pastes. |
| “No such file” for Python 3.13 | The official installer did not finish; another Python answered | Redo M2 and check with its step 4. |
| mv: … is not a directory | More than one client_secret file in Downloads | Use the newest-file command in M12 / W12. |
| Pexels: key issuance paused | Pexels stopped giving new keys | Use Pixabay (M11) or Commons only. |
| Branding Save or Publish app is grey | Google wants more branding for production apps | Use Testing mode with yourself as a test user (C4). |
| Google: “Access blocked” | Your account is not a test user | Add it in C4, then connect again. |
| RECONNECT NEEDED on YouTube | Testing mode ends the connection after 7 days | E5: reconnect on the computer, then Retry the job. |
| Fact-check “not in the expected format” | The AI answered messily | Already handled: smaller checks, exact answer shapes, automatic second try. Tap Retry if it still stops. |
| “Only N facts could be confirmed” | The topic is poorly documented | Choose a better-documented topic and Start video. |
| Video has no captions | The caption font was not found | Send a screenshot; it is a quick fix. |
| Python blocked by firewall (Windows) | Windows asks before allowing network access | Allow on Private networks; set your Wi-Fi to Private. |

# Part H: Security and backups

## What is secret, and where it lives

| Secret | Where | Protection |
|---|---|---|
| Gemini key, Pixabay key, password hash, session key | VideoFactory/.env | Only your user can read it (Mac). Never in code, logs or the dashboard. |
| YouTube client file and permission | VideoFactory/secrets/ | Locked folder (Mac). Never in the database, logs or backups. |
| Dashboard password | Only a scrambled hash is stored | 5 wrong tries locks a device out for 5 minutes. |

- Never share .env, the secrets folder, or anything starting with AIza or client_secret.
- If a key may have leaked: delete it on the provider's website, create a new one, and save it with set_key.
- Change the dashboard password any time with set_password (it logs every device out).
- The factory is not on the public internet. Only your Tailscale devices and your home Wi-Fi can reach it, and the password still applies.

## Back up your factory

A backup contains your database, scripts, research and videos (the data folder). It deliberately excludes keys and permissions, which you can always recreate. Store backups on an external drive or private cloud storage, never anywhere public.

**Mac:**

```
bash ~/VideoFactory/factory.sh stop && cd ~ && zip -r "VideoFactory-backup-$(date +%Y-%m-%d).zip" VideoFactory/data -x "VideoFactory/data/cache/*" && bash ~/VideoFactory/factory.sh start
```

**Windows:**

```
taskkill /IM waitress-serve.exe /F
Compress-Archive -Path "$HOME\VideoFactory\data" -DestinationPath "$HOME\VideoFactory-backup-$(Get-Date -Format yyyy-MM-dd).zip" -Force
Start-ScheduledTask -TaskName "VideoFactory"
```

## Restore

1. Set up the factory with Part A or B (steps up to M8 or W8).
2. Stop the factory, then unpack the backup into your home folder so the data folder lands inside VideoFactory. Mac: cd ~ && unzip -o ~/Downloads/VideoFactory-backup-DATE.zip -d ~ Windows: Expand-Archive -Path "$HOME\Downloads\VideoFactory-backup-DATE.zip" -DestinationPath "$HOME\VideoFactory" -Force
3. Save your keys again with set_key, set your password with set_password, and redo M12 / W12 and C6 for YouTube.
4. Start the factory. Your channels, topics and videos are back.

# Part I: What comes next

| Checkpoint | Status |
|---|---|
| 1 to 13: environment, software, app, database, AI, topics, script, visuals, voice, rendering, captions, quality control | Done |
| 14 and 15: YouTube connection and test upload | Built; confirm with your first private upload |
| 16: Scheduler (videos per day, posting times, automatic uploads) | Next |
| 17: Full autopilot (topic to upload with no taps) | Planned |
| 18: Analytics and learning from results | Planned |
| YouTube API audit (so uploads can be public automatically) | Planned, free, guided |
| Instagram Reels and TikTok (official APIs, require their own app approval) | Later |

# Appendix 1: Costs and limits

| Service | Used for | Cost | Limits and notes |
|---|---|---|---|
| Python, Flask, Waitress, SQLite, FFmpeg | App, dashboard, database, video | Free, open source | Runs on your computer |
| Google Gemini API (text) | Topics, research, fact-check, script, scenes, titles | Free tier | Per-minute and daily limits; factory waits and retries. Free-tier data may be used by Google: no private information. |
| Google Gemini TTS | Narration | Free tier | One request per video; limits apply |
| Wikipedia API | Research sources | Free, no key | Every source link is shown |
| Wikimedia Commons | Historical pictures | Free, no key | Only free licences accepted; credits kept |
| Pixabay API | Stock photos (optional) | Free key | About 100 requests per minute; searches cached 24 hours |
| Pexels API | Stock photos | Free | New keys paused (October 2026); used automatically if you ever add a key |
| Tailscale Personal | Permanent private link | Free | Six users, unlimited personal devices (August 2026) |
| Google Cloud + YouTube Data API | Uploading | Free | Daily allowance; uploads private until the free audit |

# Appendix 2: Folder map

| Path inside VideoFactory | What it is |
|---|---|
| app/ | The application code |
| bin/ | FFmpeg and FFprobe |
| venv/ | The private Python toolbox |
| data/factory.db | The database (the factory's memory) |
| data/jobs/NUMBER/ | Everything for one video: sources, script, pictures, audio, captions, final video, reports |
| data/logs/app.log | Activity log |
| .env | Secret keys and password hash |
| secrets/ | YouTube client file and permission |
| factory.sh, start.sh | Mac control and start scripts |
| start-windows.bat | Windows start script |

# Appendix 3: Setup checklist

| Done | Step |
|---|---|
| ☐ | Python 3.13 installed (M2 / W2) |
| ☐ | Package unpacked (M3 / W3) |
| ☐ | FFmpeg with captions (M4 / W4) |
| ☐ | Private toolbox and add-ons (M5 / W5) |
| ☐ | Gemini key saved (M6 / W6) |
| ☐ | Dashboard password (M7 / W7) |
| ☐ | System check green (M8 / W8) |
| ☐ | Automatic start (M9 / W9) |
| ☐ | Tailscale on computer and phone (M10 / W10, D1) |
| ☐ | Pixabay key (optional, M11 / W11) |
| ☐ | Google Cloud setup (C1 to C5) |
| ☐ | YouTube connected (M12 / W12, C6) |
| ☐ | First channel, topic and video (E1 to E3) |
| ☐ | First private YouTube upload (E4) |